

FIXED POINTS OF SYMPLECTOMORPHISMS OF S^2

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ABSTRACT. This document is a set of notes written in preparation for a talk given by the author for MATH 269 (Topics in Symplectic Geometry), taught by Mohammed Abouzaid in Spring 2025.

Outline

- (1) Introduction to the problem: Nikishin's theorem for symplectomorphisms on S^2
- (2) Warmup: symplectomorphisms close to the identity
- (3) Obstruction for a diffeomorphism with 2 fixed points: index 2 fixed point
- (4) Nondegenerate case: Morse!
- (5) Degenerate case: reduction to understanding singularity
- (6) Final Morse theory lemma

1. INTRODUCTION

Let us equip S^2 with the symplectic form ω given by its usual volume form. My goal today is to prove the following theorem, which is the degenerate Arnold conjecture for arbitrary symplectomorphisms on S^2 and (to my knowledge) the first proof of a result confirming Arnold's conjecture.

Theorem 1.1 (Nikishin 1972). *Let $f: S^2 \rightarrow S^2$ be a symplectomorphism. Then $\# \text{Fix } f \geq 2$.*

Remark 1.2. Of course, the minimum number of critical points of a smooth function on S^2 is 2: the lower bound holds because S^2 is compact and the upper bound is attained by the height function.

I will start by proving several weaker versions of the theorem which apply in greater generality, then eventually I will present Nikishin's argument.

2. SYMPLECTOMORPHISMS C^1 -CLOSE TO THE IDENTITY

The first version, as a bit of a warmup, has to do with symplectomorphisms which are sufficiently C^1 -close to the identity.

Let (M, ω) be a closed symplectic manifold and let $f: M \xrightarrow{\sim} M$ be a symplectomorphism. Then, note that the graph

$$\Gamma_f \subseteq (M \times M, \omega \oplus (-\omega))$$

is a Lagrangian submanifold:

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Proposition 2.1. *Let $f: M \rightarrow M$ be a diffeomorphism of a symplectic manifold (M, ω) . Then,*

$$\text{graph } f \subseteq (M \times M, \tilde{\omega}) \text{ is Lagrangian} \iff f \text{ is a symplectomorphism.}$$

Proof. Recall that a submanifold is Lagrangian iff it is half dimensional and the pullback of the symplectic form under the inclusion is zero. Let $\iota: M \rightarrow M \times M$ be the inclusion of the graph;

$$\iota^* \tilde{\omega} = \iota^* \pi_1^* \omega - \iota^* \pi_2^* \omega = \omega - f^* \omega.$$

Hence the result follows. $\square$

Now recall *Weinstein's tubular neighborhood theorem*, which says that any compact Lagrangian L in a symplectic manifold has a tubular neighborhood which is symplectomorphic to a neighborhood of the zero section in T^*L . Now let us apply this to the Lagrangian $\Delta \subseteq M \times M$, so that we get a neighborhood $U \supseteq \Delta$ and $U_0 \supseteq \Delta_0 \subseteq T^*\Delta$ and a symplectomorphism $\Phi: U \rightarrow U_0$ satisfying $\Phi(p, p) = (p, 0)$.

Define $\iota: M \rightarrow M \times M$ the embedding of the diagonal and $j: M \rightarrow M \times M$ the embedding of the graph. Since f is C^1 -close to id, we can assume that $\Gamma_f \subseteq U$. Let $\iota_0 = \varphi \circ \iota$ and $j_0 = \varphi \circ j$ be the induced embeddings into $T^*\Delta$. Assuming f is sufficiently C^1 -close to the identity, the projection $\pi: T^*\Delta \rightarrow \Delta$ when restricted to Γ_f is injective, so it follows that $\Gamma_f \subseteq T^*\Delta$ is also the graph of a 1-form $\alpha \in \Omega^1(\Delta) = \Omega^1(M)$.

Finally, the fact that $\Gamma_f \subseteq M \times M$ is Lagrangian gives us that $\Gamma_\alpha \subseteq T^*\Delta$ is Lagrangian and hence α is closed:

Proposition 2.2. *Let $\alpha \in \Omega^1(X)$ be a 1-form. Then, the graph*

$$\text{graph } \alpha = \{(x, \alpha_x) \in T^*X : x \in X\}$$

is Lagrangian if and only if α is closed.

Proof. Denote $\iota: X \hookrightarrow T^*X$ the embedding of α , considered as a section. Let λ denote the tautological 1-form on T^*X , so that $d\lambda = \omega$. Then,

$$\iota^* \omega = \iota^* d\lambda = d\iota^* \lambda = d\alpha,$$

by definition of the tautological 1-form, so $\text{graph } \alpha$ is Lagrangian if and only if α is closed. $\square$

To check the formula $\iota^* \lambda = \alpha$, we check pointwise: for $p = (x, \xi) \in T^*X$ recall that λ is defined to be $\lambda_p = (d\pi_p)^* \xi$. Then, for $p = \iota(x) = (x, \alpha_x)$,

$$(\iota^* \lambda)_x = (d\iota_x)^* \lambda_p = (d\iota_x)^* (d\pi_p)^* \alpha_x = (d(\iota \circ \pi)_p)^* \alpha_x = \alpha_x$$

since ι is a section.

To recap: we've started with a symplectomorphism $f: M \rightarrow M$ and ended up with a closed 1-form $\alpha \in \Omega^1(M)$ such that $\Gamma_f \cong \Gamma_\alpha$ under the isomorphism in the Weinstein tubular neighborhood theorem. We would now like to connect ourselves to Morse theory; hence we would like a smooth function on M . The natural candidate for this is to try to look for a primitive for α .

Theorem 2.3. *Let (M, ω) be a closed symplectic manifold with $H^1(M) = 0$ and $f: M \xrightarrow{\sim} M$ a symplectomorphism that is C^1 -close to the identity. Then $\# \text{Fix } f \geq 2$.*

Proof. Since $H^1(M) = 0$, by above we can find a smooth function $g \in C^\infty(M)$ such that $dg = \alpha$. But, note that critical points of g exactly correspond to zeroes of α , which are intersection points of Γ_α with Δ_0 the zero section. But, under the symplectomorphism $\Phi: U \rightarrow U_0$ these intersection points are precisely the intersection points $\Gamma_f \cap \Delta$, which are exactly the fixed points of f .

Since M is compact, g has at least two critical points, and hence $\# \text{Fix } f \geq 2$. $\square$

Remark 2.4. Here, the function g is called a *generating function* for the symplectomorphism f . Notice that we did not assume g was Morse: in fact, this generating function can in general be quite bad: it may not have isolated critical points, for instance. However, since we are seeking a lower bound for the number of critical points, we actually don't need to care about any of this: henceforth, we will always assume our critical points are isolated (otherwise there are infinitely many and the theorem is proved).

Hence, we have now proved Nikishin's theorem in the case where f is sufficiently close to the identity. But, in some sense we have traded global power for local power: in restricting our symplectomorphisms to those close to id, we were able to use a Morse-theoretic fact about all of M . We will now aim to use a somewhat opposite strategy: we allow ourselves to take arbitrary symplectomorphisms, but in exchange we have to apply our Morse theoretic tools locally.

3. MOTIVATION FOR THE FAR CASE

We recall the Lefschetz fixed point theorem.

Let $\varphi: X \xrightarrow{\sim} X$ be a diffeomorphism of a smooth manifold X^n , and $p \in X$ a fixed point. We define the *index* of p as follows: fix a chart (U, ψ) at p (so that $\psi(p) = 0$), and consider the map $V = \psi \circ \varphi \circ \psi^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$. The *index* of p is the index of the vector field V at 0.

Let us define the index of an isolated fixed point as follows. Let $f: X \rightarrow X$ be a diffeomorphism and $p \in \text{Fix}(f)$ isolated. Take a chart (U, φ) about p containing no other fixed points in $\overline{U}$, and write $q = \varphi(p)$, and let $r > 0$ be small enough such that $B_r(q) \subseteq \varphi(U)$ and moreover that $f(\varphi^{-1}(B_r(q))) \subseteq U$. On this set $B = B_r(q)$, we can define the vector field $V \in C^\infty(B, \mathbb{R}^n)$

$$V(x) = (\varphi \circ f \circ \varphi^{-1})(x) - x.$$

Of course, $\varphi(f(\varphi^{-1}(q))) = \varphi(f(p)) = \varphi(p) = q$, so $V(q) = 0$. Moreover, since p is an isolated fixed point (and φ is a diffeomorphism), q is an isolated zero of V , and the unique zero on B .

Then, as usual we define

$$\text{index}_B V = \deg \left(\frac{V}{|V|}: \partial B \rightarrow S^{n-1} \right),$$

and using this we define

$$\text{index}_f(p) = \text{index}_B V.$$

The usual worry is that this is not well-defined: there are two things to check. First we need to check that the index is independent of the choice of $r > 0$, and then we need to check that the index is independent of the choice of chart.

The first is not too hard by the usual arguments about the degree of a smooth map which extends to the interior. The second is straightforward, but we just have to keep track of orientations. The point is that the transition functions are both diffeomorphisms and hence change the degree by a multiplicative factor of ± 1 , but this is done twice and hence cancels out.

Theorem 3.1. *Let X be a smooth manifold with $\varphi: X \rightarrow X$ a diffeomorphism. Then,*

$$\sum_{p \in \text{Fix}(\varphi)} \text{index}_\varphi(p) = \sum_{k=0}^n (-1)^k \text{tr}(\varphi_*: H_k(X; \mathbb{Q}) \rightarrow H_k(X; \mathbb{Q})).$$

In the case of a symplectomorphism $f: S^2 \rightarrow S^2$, f preserves orientation and hence the Lefschetz number is 2. But, note that simply preserving orientation is not sufficient for Nikishin's theorem to hold: let us consider the following example.

Example 3.2. Consider $S^2 = \mathbb{C} \cup \{\infty\}$ as the Riemann sphere and take the projective linear transformation

$$\varphi(z) = z + b$$

for $b \neq 0$. Here, ∞ is a fixed point. To see this more rigorously, we can change coordinates to $w = \frac{1}{z}$ and note that on the overlap of the two charts $\{z, w \neq 0\}$, we can write

$$\varphi(w) = \frac{1}{w^{-1} + b} = \frac{w}{1 + bw}.$$

Then, we can see that $w = 0$ is a fixed point. Moreover, note that near $w = 0$ we have the Taylor expansion

$$\varphi(w) = w(1 - bw + b^2w^2 - \dots) = w - bw^2 + b^2w^3 - \dots,$$

so

$$\varphi(w) - w = -bw^2 + \dots$$

and hence the index of this fixed point is 2.

The above example, along with the Lefschetz fixed point theorem, tells us that the only obstruction to an orientation-preserving diffeomorphism of the sphere having at least 2 fixed points is the existence of an index 2 fixed point. So, we can say: if $f: S^2 \rightarrow S^2$ is an orientation-preserving diffeomorphism with isolated fixed points, each of index $\neq 2$, then there must be at least two fixed points.

4. NONDEGENERATE FIXED POINTS

One easy way to ensure that fixed points have index $\neq 2$ is to ensure that they are nondegenerate: i.e., the generating function is Morse. But, now that we're looking at an arbitrary symplectomorphism, we have no way of ensuring that Γ_f can be written as the graph of a 1-form. But, we can always do so locally near fixed points, giving us the notion of *local generating functions*. This

has the additional benefit of being potentially useful in cases with $H^1 \neq 0$, since closed forms are always locally exact.

This is a purely smooth notion (no reference to the symplectic structure or the symplectomorphism).

Let X^n be a smooth manifold and let $f: X \rightarrow X$ be a diffeomorphism. Denote

$$\Delta = \text{graph id} = \{(x, x) \in X \times X : x \in X\},$$

$$\Gamma_f = \text{graph } f = \{(x, f(x)) \in X \times X : x \in X\},$$

and note that the set of fixed points $\text{Fix}(f)$ is in one-to-one correspondence with $\Delta \cap \Gamma_f$. We say that a fixed point $p \in \text{Fix}(f)$ is *nondegenerate* if the intersection $(p, p) \in \Delta \cap \Gamma_f$ is transverse.

We have an equivalent characterization in terms of df_p . Let $\iota: X \rightarrow X \times X$ denote the embedding $\iota: x \mapsto (x, f(x))$. For $p \in \text{Fix}(f)$, note that $d\iota_p: T_p X \rightarrow T_p X \times T_p X$ is given by $d\iota_p = (\text{id}_{T_p X}, df_p)$. So, the intersection is transverse if and only if $\text{im } df_p \cap \Delta_{T_p X} = \{0\}$, if and only if the only solutions to $df_p(\xi) = \xi$ is $\xi = 0$, if and only if df_p does not have 1 as an eigenvalue if and only if $df_p - \text{id}_{T_p X}$ is invertible.

If the intersection is not transverse, then we call p a *degenerate* fixed point.

Now assume $f: X \rightarrow X$ is a diffeomorphism with isolated nondegenerate fixed points. It turns out that this is equivalent to the fixed points having index ± 1 .

Let $p \in \text{Fix } f$, and let (U, φ) be a chart at p and $V = (\varphi \circ f \circ \varphi^{-1}) - \text{id}$ a vector field defined on a ball B near $\varphi(p)$.

Then, p is nondegenerate if and only if $df_p - \text{id}_{T_p X}: T_p X \rightarrow T_p X$ is invertible; but, expressing this map in coordinates we get that $dV_0: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible:

Computing dV_q , we have

$$dV_q(\xi) = d\varphi_p \circ df_p \circ d\varphi_q^{-1} \xi - \xi.$$

Since dV_q is a map between vector spaces of the same dimension, it is bijective if and only if it is injective. Note, then, that the kernel of dV_q is

$$\ker dV_q = \{\xi \in \mathbb{R}^n : d\varphi_p \circ df_p \circ d\varphi_q^{-1} \xi = \xi\} = d\varphi_p(\{\zeta \in \mathbb{R}^n : df_p \zeta = \zeta\})$$

by taking $\zeta = d\varphi_q^{-1}(\xi)$, since φ is a diffeomorphism onto its image. Hence

$$\dim \ker dV_q = \dim \ker(df_p - \text{id}_{T_p X}),$$

so once again dV_q is an isomorphism if and only if p is a nondegenerate fixed point.

Hence in this case, where $dV_0: T_0 B \rightarrow T_0 \mathbb{R}^n$ is an isomorphism, we can reduce to the linearization.

Proposition 4.1. *Let $V \in C^\infty(B_r(q), \mathbb{R}^n)$ be a vector field with a unique zero at q and $V \neq 0$ on $\overline{B} = \overline{B_r(q)}$. If $dV_0: T_0 B \rightarrow T_0 \mathbb{R}^n$ is an isomorphism, then*

$$\text{index}_B V = \deg \left(\frac{dV_q}{|dV_q|}: \partial B \rightarrow S^{n-1} \right) = \text{sgn } \det dV_q,$$

considered as a map $dV_q: \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Hence in the case that p is a nondegenerate fixed point, we have

$$\text{index}_f(p) = \text{sgn det } dV_q = \pm 1,$$

as desired.

So, we conclude: if $f: X \rightarrow X$ is a diffeomorphism with nondegenerate fixed points, then $\# \text{Fix } f \geq |\Lambda_f|$.

5. THE DEGENERATE CASE

It follows that the hard part about proving Nikishin's theorem (or rather, the part that relies on the symplectic machinery) is the case where f has degenerate fixed points. If we were allowed to globally perturb f to get rid of the degenerate fixed points this would be simple, but because we want to keep our specific symplectomorphism f this is not possible. But, note that in analyzing each (isolated) fixed point locally, we can allow ourselves a little bit of flexibility to get quantitative controls on the index of the fixed point, once we reduce it to the degree of a certain map associated to a vector field.

Let $f: S^2 \rightarrow S^2$ be a symplectomorphism, and let $x_0 \in \text{Fix } f$ be an isolated fixed point. Then, using Darboux's theorem we can find symplectic coordinates (p, q) on the source and (P, Q) on the target such that the symplectic form in coordinates is $dp \wedge dq$ and $dP \wedge dQ$, respectively. Write $f(p, q) = (f^P(p, q), f^Q(p, q))$, where $f^P = P \circ f$ and $f^Q = Q \circ f$. The symplectomorphism condition in these coordinates can be written as

$$\det df = \det \begin{bmatrix} \frac{\partial f^P}{\partial p} & \frac{\partial f^P}{\partial q} \\ \frac{\partial f^Q}{\partial p} & \frac{\partial f^Q}{\partial q} \end{bmatrix} = 1.$$

Cutting off by a smooth function if necessary, let us reduce to the case of a symplectomorphism $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with an isolated fixed point at 0, and denote $p, q: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $P, Q: \mathbb{R}^2 \rightarrow \mathbb{R}$ the coordinate functions on the source and target.

In these coordinates, we have $V(p, q) = (f^P(p, q) - p, f^Q(p, q) - q)$, and hence as above we have the equivalent characterizations

$$0 \text{ nondegenerate fixed point of } f \iff df_0 - I = dV_0 \text{ invertible.}$$

In coordinates,

$$\det dV_0 = (\partial_p f^P(0) - 1)(\partial_q f^Q(0) - 1) - \partial_q f^P(0)\partial_p f^Q(0) = \det df_0 - \partial_p f^P(0) - \partial_q f^Q(0) + 1 = 2 - \text{tr } df_0.$$

Hence in the degenerate case, we have $\text{tr } df_0 = 2$.

At this point I'd like to look back at the picture. The fact that 0 is a degenerate fixed point means that the intersection point $0 \in \Gamma_f \cap \Delta$ is not transverse. This is a kind of polarization-independent fact. But, when we compute the index of the fixed point, we implicitly identify the ambient $\mathbb{R}^4$ with a choice of a Lagrangian affine subspace $\mathbb{R}^2$ and complementary space $\mathbb{R}^2$ to make $\mathbb{R}^4 \cong \mathbb{R}^2 \times \mathbb{R}^2 \cong T^*\mathbb{R}^2$. In the spirit of the earlier fact that the index of the fixed point doesn't depend on the choice of chart, as long as we keep our orientations straight, we have a similar result: we can choose a different polarization of $\mathbb{R}^4$, and as long as the submanifolds Γ_f and Δ remain Lagrangian and the relevant transition maps are orientation preserving diffeomorphisms on both spaces, we can use this new coordinate system to compute the index.

From the computation above we have $\det dV_0 = 0$ and hence $\text{tr } df_0 = 2 > 0$. Hence at least one of $\partial_p f^P$ or $\partial_q f^Q$ is positive; without loss of generality assume $\partial_p f^P(0) > 0$. What this means in particular is as follows: let $\pi_{p,q} = (p, q): \mathbb{R}_{p,q,P,Q}^4 \rightarrow \mathbb{R}_{p,q}^2$ denote the projection onto the first two coordinates, and $\pi_{P,q} = (P, q): \mathbb{R}^4 \rightarrow \mathbb{R}^2$ denote the projection onto the first and third coordinates.

Notice that $\Delta \subseteq \mathbb{R}^4$ is a graph over $\mathbb{R}_{p,q}^2$ of the function $(P, q) \mapsto (p, q) \in \mathbb{R}_{p,q}^2$. Hence the maps

$$\mathbb{R}_{p,q}^2 \xleftarrow{\pi_{p,q}} \Delta \xrightarrow{\pi_{P,q}} \mathbb{R}_{P,q}^2$$

are all smooth diffeomorphisms (globally, but also in particular near 0).

We would like to have the same fact be true of Γ_f . But, this follows from the implicit function theorem: let $F: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ be the local defining function

$$F(p, q, P, Q) = f(p, q) - (P, Q).$$

In order for the projection onto the (P, q) -plane to be a diffeomorphism near 0, we need to have

$$\det \left(\frac{\partial(f^P, q)}{\partial(p, q)} \right) \neq 0.$$

Computing the full matrix dF_0 , we have

$$dF_0 = \begin{bmatrix} \frac{\partial f^P}{\partial p} & \frac{\partial f^P}{\partial q} & -1 & 0 \\ \frac{\partial f^Q}{\partial p} & \frac{\partial f^Q}{\partial q} & 0 & -1 \end{bmatrix}$$

In order to have the projection to (P, q) coordinates be a diffeomorphism near 0, we need the outer two columns to be invertible. But, the determinant is simply $-\partial_p f^P < 0$, so our map is orientation reversing. But, notice that if we put in $f = \text{id}$ into this matrix, we get $(I, -I)$, so the same change of coordinates for Δ is also orientation reversing; hence, the index stays the same.

Hence Γ_f is graphical over the (P, q) -plane.

Above, when considering the index of the fixed point we took Γ_f as graphical over (p, q) to (P, Q) and took a neighborhood of Δ as $T^*\Delta$. Now, we preserve the second part and change the first part.

Now take Γ_f as graphical over the (P, q) -plane, with the function $\tilde{f}(P, q)$. Notice that the symplectic form on $\mathbb{R}^4$ can be written as $dp \wedge dq - dP \wedge dQ = d(p dq + Q dP)$. That Γ_f is Lagrangian implies that the pullback of this form is 0; hence $\tilde{f}^P dq + \tilde{f}^Q dP$ (considered as a form on $\mathbb{R}_{P,q}^2$) is closed. Since $H^1(\mathbb{R}^2) = 0$, we can find a primitive $g \in C^\infty(\mathbb{R}_{P,q}^2)$ such that

$$(*) \quad dg = \tilde{f}^P dq + \tilde{f}^Q dP.$$

Since we are free to choose g up to addition by a constant, we can stipulate that $g(0, 0) = 0$.

Now define the function $h(P, q)$ by

$$h(P, q) = g(P, q) - Pq.$$

Clearly $h(P, q)$ is smooth and vanishes at 0. From the equation (*) we get that

$$dg = \left(\frac{\partial h}{\partial q} + P \right) dq + \left(\frac{\partial h}{\partial P} + q \right) dP,$$

and hence

$$\tilde{f}^P(P, q) - P = -\frac{\partial h}{\partial P} \quad \text{and} \quad \tilde{f}^Q(P, q) - q = -\frac{\partial h}{\partial q}.$$

The final upshot of all of this is that if we define

$$\tilde{V}(P, q) = (\tilde{f}^P(P, q) - P, \tilde{f}^Q(P, q) - q),$$

then we have that

$$\tilde{V} = -\nabla h,$$

so the vector field whose index at 0 we would like to compute is actually the gradient of a smooth function.

6. THE FINAL MORSE THEORY LEMMA

At this point, the result follows from the following lemma.

Lemma 6.1. *Let $B \subseteq \mathbb{R}^2$ be an open ball with $f \in C^\infty(\overline{B})$ function with a unique isolated critical point at 0 and $V = \nabla f \neq 0$ on ∂B . Then, there exists a small C^∞ perturbation $\tilde{f}$ such that $\tilde{f}$ is Morse and $\tilde{f}$ has at most one critical point of Morse index 0 or 2.*

The idea is that if we had a local min and a local max, the birth-death picture would have to give us a sphere, which is a contradiction. If we had two local minima, we must have a Morse index 1 critical point and can perturb them away.

As a corollary, since Morse index 1 critical points contribute -1 to the topological index and Morse index 0, 2 critical points contribute $+1$ to the topological index, it follows that the index can be at most 1, which proves our result.

Nikishin proves this result with a different method from ODE, by analyzing the stability of a stationary point of an ODE. The idea is as follows: suppose that we have a time-independent ODE

$$x'(t) = F(x) \quad \text{and} \quad x(0) = x_0.$$

Then, near a stationary point a (where $F(a) = 0$), on a small enough ball the behavior of the system acts roughly according to the linearized system $x'(t) = dF(x)$, and we can analyze the system according to the eigenvalues of dF_a . In our case, since $F = \nabla h$, $dF = \text{Hess } h$ is symmetric, it has only real eigenvalues and is diagonalizable. Imaginary eigenvalues correspond to rotation behavior, so this rules out rotation. Non-diagonalizability makes us need to use the Jordan block, which gives us degenerate behavior. Hence we are in very nice territory.

What this means is that in a sufficiently small ball, the solutions to the ODE $x' = \nabla h$ are either a center (sink/source) or have $2k$ hyperbolic sectors, which correspond to the stable/unstable manifolds of this critical point under gradient flow. Looking at the structure of such a critical point, we conclude that the index is $1 - k$. One way to see this is by looking at the degree and winding around the boundary.

Another way to see this is to notice that a stationary point with $2k$ hyperbolic sectors, after Morsification, splits into $k - 1$ saddle points, each of which contribute -1 to the topological index.